MATH 111 Calculus 1

Continuity, part II

Building continuous functions · The Intermediate Value Theorem

Week 2 · Lecture 5 · Stewart §1.8

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Sit with your team. In the meantime: is $\sqrt{x^2+1}$ continuous? Where? And $\sin(1/x)$?

Two questions from Lecture 4

"Is $f(x) = 1/x$ a continuous function or not?"

Two questions from Lecture 4

"Is $f(x) = 1/x$ a continuous function or not?"

Yes — at every number in its domain, which is what "continuous function" means. At 0 the function is not defined, so there is nothing to check there; the graph of a function is allowed to come in two pieces. "Continuous on an interval" is the statement with the unbroken curve.

Two questions from Lecture 4

"Is $f(x) = 1/x$ a continuous function or not?"

Yes — at every number in its domain, which is what "continuous function" means. At 0 the function is not defined, so there is nothing to check there; the graph of a function is allowed to come in two pieces. "Continuous on an interval" is the statement with the unbroken curve.

"If a function has a hole, is it discontinuous even though we never use that point?"

Two questions from Lecture 4

"Is $f(x) = 1/x$ a continuous function or not?"

Yes — at every number in its domain, which is what "continuous function" means. At 0 the function is not defined, so there is nothing to check there; the graph of a function is allowed to come in two pieces. "Continuous on an interval" is the statement with the unbroken curve.

"If a function has a hole, is it discontinuous even though we never use that point?"

At that point, yes — and the discontinuity is *removable*: define $f(a)$ to be the limit and the function becomes continuous. That is exactly what we did in Lecture 2 when we cancelled $(x-2)$ in $\frac{x^2-4}{x-2}$ and got $x+2$.

Building continuous functions: sums, products, quotients

Theorem

If f and g are continuous at a and c is a constant, then so are $f+g$, $f-g$, cf, fg, and f/g provided $g(a) \neq 0$.

Building continuous functions: sums, products, quotients

Theorem

If f and g are continuous at a and c is a constant, then so are $f+g$, $f-g$, cf, fg, and f/g provided $g(a) \neq 0$.

Why — the sum, in two lines (the others are the same):

Building continuous functions: sums, products, quotients

Theorem

If f and g are continuous at a and c is a constant, then so are $f+g$, $f-g$, cf, fg, and f/g provided $g(a) \neq 0$.

Why — the sum, in two lines (the others are the same):

$$\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x) \quad \text{(sum law: both limits exist)}$$

Building continuous functions: sums, products, quotients

Theorem

If f and g are continuous at a and c is a constant, then so are $f+g$, $f-g$, cf, fg, and f/g provided $g(a) \neq 0$.

Why — the sum, in two lines (the others are the same):

$$\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) \quad \text{(sum law: both limits exist)}$$

$$= f(a) + g(a) = (f+g)(a) \quad \text{(continuity of } f \text{ and } g \text{ at } a\text{)}.$$

Building continuous functions: sums, products, quotients

Theorem

If f and g are continuous at a and c is a constant, then so are $f+g$, $f-g$, cf, fg, and f/g provided $g(a) \neq 0$.

Why — the sum, in two lines (the others are the same):

$$\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) \quad \text{(sum law: both limits exist)}$$
$$= f(a)+g(a) = (f+g)(a) \quad \text{(continuity of } f \text{ and } g \text{ at } a\text{)}.$$

So $f+g$ is continuous at a. For the quotient, the quotient law needs $\lim g(x) = g(a) \neq 0$ — that is where the condition comes from.

Building continuous functions: sums, products, quotients

Theorem

If f and g are continuous at a and c is a constant, then so are $f+g$, $f-g$, cf, fg, and f/g provided $g(a) \neq 0$.

Why — the sum, in two lines (the others are the same):

$$\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) \quad \text{(sum law: both limits exist)}$$

$$= f(a) + g(a) = (f+g)(a) \quad \text{(continuity of } f \text{ and } g \text{ at } a\text{)}.$$

So $f+g$ is continuous at a. For the quotient, the quotient law needs $\lim g(x) = g(a) \neq 0$ — that is where the condition comes from.

Consequence. x and the constants are continuous; multiply and add: every polynomial is continuous on $\mathbb{R}$; divide: every rational function is continuous on its domain — yesterday's theorem, now with a proof.

Building continuous functions: composition

Theorem

If f is continuous at b **and** $\lim_{x \to a} g(x) = b$, **then**

$$\lim_{x \to a} f\big(g(x)\big) \;=\; f(b) \;=\; f\Big(\lim_{x \to a} g(x)\Big).$$

Building continuous functions: composition

Theorem

If f is continuous at b **and** $\lim_{x \to a} g(x) = b$, **then**

$$\lim_{x \to a} f\big(g(x)\big) \;=\; f(b) \;=\; f\Big(\lim_{x \to a} g(x)\Big).$$

In words: a limit may be moved inside a continuous function. And if g is continuous at a and f is continuous at $g(a)$, then $f \circ g$ is continuous at a.

Building continuous functions: composition

Theorem

If f is continuous at b **and** $\lim_{x \to a} g(x) = b$, **then**

$$\lim_{x \to a} f\big(g(x)\big) = f(b) = f\Big(\lim_{x \to a} g(x)\Big).$$

In words: a limit may be moved inside a continuous function. And if g is continuous at a and f is continuous at $g(a)$, then $f \circ g$ is continuous at a.

Example. $h(x) = \sin(x^2)$ is continuous on $\mathbb{R}$: x^2 is a polynomial, sin is continuous everywhere.

Building continuous functions: composition

Theorem

If f is continuous at b **and** $\lim_{x \to a} g(x) = b$, **then**

$$\lim_{x \to a} f\big(g(x)\big) = f(b) = f\Big(\lim_{x \to a} g(x)\Big).$$

In words: a limit may be moved inside a continuous function. And if g is continuous at a and f is continuous at $g(a)$, then $f \circ g$ is continuous at a.

Example. $h(x) = \sin(x^2)$ is continuous on $\mathbb{R}$: x^2 is a polynomial, sin is continuous everywhere.

Example. Where is $F(x) = \dfrac{1}{\sqrt{x^2+7}-4}$ continuous?

Building continuous functions: composition

Theorem

If f is continuous at b **and** $\lim_{x\to a} g(x) = b$, **then**

$$\lim_{x\to a} f\big(g(x)\big) \;=\; f(b) \;=\; f\Big(\lim_{x\to a} g(x)\Big).$$

In words: a limit may be moved inside a continuous function. And if g is continuous at a and f is continuous at $g(a)$, then $f \circ g$ is continuous at a.

Example. $h(x) = \sin(x^2)$ is continuous on $\mathbb{R}$: x^2 is a polynomial, sin is continuous everywhere.

Example. Where is $F(x) = \dfrac{1}{\sqrt{x^2+7}-4}$ continuous? Built in continuous steps (x^2+7, $\sqrt{\ }$, -4, $1/$); the only danger is the denominator: $\sqrt{x^2+7} = 4 \iff x = \pm 3$.

Building continuous functions: composition

Theorem

If f is continuous at b **and** $\lim_{x\to a} g(x) = b$, **then**

$$\lim_{x\to a} f\big(g(x)\big) \;=\; f(b) \;=\; f\Big(\lim_{x\to a} g(x)\Big).$$

In words: a limit may be moved inside a continuous function. And if g is continuous at a and f is continuous at $g(a)$, then $f \circ g$ is continuous at a.

Example. $h(x) = \sin(x^2)$ is continuous on $\mathbb{R}$: x^2 is a polynomial, sin is continuous everywhere.

Example. Where is $F(x) = \dfrac{1}{\sqrt{x^2+7}-4}$ continuous? Built in continuous steps (x^2+7, $\sqrt{\ }$, -4, $1/$); the only danger is the denominator: $\sqrt{x^2+7} = 4 \iff x = \pm 3$.

So F is continuous everywhere except at ± 3: on $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$.

Two quick uses

Evaluate $\lim_{x \to \pi} \sin\big(x + \sin x\big)$.

Two quick uses

Evaluate $\lim_{x \to \pi} \sin\big(x + \sin x\big)$.

sin is continuous, so the limit moves inside: $\lim_{x \to \pi} \sin(x + \sin x) = \sin\Big(\lim_{x \to \pi} (x + \sin x)\Big)$

Two quick uses

Evaluate $\lim_{x \to \pi} \sin\big(x + \sin x\big)$.

sin is continuous, so the limit moves inside: $\lim_{x \to \pi} \sin(x + \sin x) = \sin\Big(\lim_{x \to \pi} (x + \sin x)\Big)$

$x + \sin x$ is continuous (sum of continuous functions), so substitute:
$= \sin(\pi + \sin \pi) = \sin(\pi + 0) = 0$.

Two quick uses

Evaluate $\lim_{x \to \pi} \sin\big(x + \sin x\big)$.

sin is continuous, so the limit moves inside: $\lim_{x \to \pi} \sin(x + \sin x) = \sin\Big(\lim_{x \to \pi} (x + \sin x)\Big)$

$x + \sin x$ is continuous (sum of continuous functions), so substitute:
$= \sin(\pi + \sin \pi) = \sin(\pi + 0) = 0$.

Where is $f(x) = \dfrac{\sqrt{x}}{x^2 - 1}$ continuous?

Two quick uses

Evaluate $\lim_{x \to \pi} \sin\big(x + \sin x\big)$.

sin is continuous, so the limit moves inside: $\lim_{x \to \pi} \sin(x + \sin x) = \sin\Big(\lim_{x \to \pi} (x + \sin x)\Big)$

$x + \sin x$ is continuous (sum of continuous functions), so substitute:
$= \sin(\pi + \sin \pi) = \sin(\pi + 0) = 0$.

Where is $f(x) = \dfrac{\sqrt{x}}{x^2 - 1}$ continuous?

$\sqrt{x}$ needs $x \geq 0$; the quotient needs $x^2 \neq 1$, i.e. $x \neq \pm 1$. Root, polynomial, quotient — continuous wherever defined.

Two quick uses

Evaluate $\lim_{x \to \pi} \sin\big(x + \sin x\big)$.

sin is continuous, so the limit moves inside: $\lim_{x \to \pi} \sin(x + \sin x) = \sin\Big(\lim_{x \to \pi} (x + \sin x)\Big)$

$x + \sin x$ is continuous (sum of continuous functions), so substitute:
$= \sin(\pi + \sin \pi) = \sin(\pi + 0) = 0$.

Where is $f(x) = \dfrac{\sqrt{x}}{x^2 - 1}$ continuous?

$\sqrt{x}$ needs $x \geq 0$; the quotient needs $x^2 \neq 1$, i.e. $x \neq \pm 1$. Root, polynomial, quotient — continuous wherever defined.

Answer: on its domain, $[0, 1) \cup (1, \infty)$ — continuous on its domain is the phrase; at 1 there is an infinite discontinuity, and for $x < 0$ nothing at all.

Question 10

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 2 minutes

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

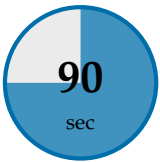


In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 2 minutes

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:



A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

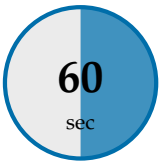
D. continuous from the right, but not from the left.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 2 minutes

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:



A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

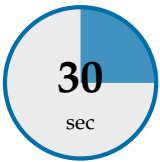
In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

Question 10

THINK & TALK · 2 minutes

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:



A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 2 minutes

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

Question 10

VOTE

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

Question 10

VOTE

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B **C** D

VOTE

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

Hands down. Thank you.

A B C D

Question 10

EXPLAIN

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

EXPLAIN

Let $h(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ At $x = 0$ the function h is:

A. continuous — sin and $1/x$ are continuous, so their composition is.

B. not continuous — $\lim_{x \to 0} h(x)$ does not exist.

C. not continuous — an infinite discontinuity.

D. continuous from the right, but not from the left.

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

B. As $x \to 0$, $1/x$ runs off to $\pm\infty$ and $\sin(1/x)$ keeps sweeping through $[-1, 1]$ — the values never settle (Lecture 1's trap), so the limit does not exist and no choice of $h(0)$ can help. A is the trap here: the composition theorem needs $1/x$ to be continuous *at* 0, and $1/x$ is not even defined there. (Not C: the values stay between -1 and 1.)

STAND UP

$$\text{“}k(x) = \begin{cases} x \sin \dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases} \text{ is continuous at } 0.\text{”}$$

Stand up if you think this is TRUE.

Stay standing until I say “sit”. Then find someone with the other opinion — 60 seconds.

STAND UP

$$\text{“}k(x) = \begin{cases} x \sin \dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases} \text{ is continuous at } 0.\text{”}$$

Stand up if you think this is TRUE.

Stay standing until I say “sit”. Then find someone with the other opinion — 60 seconds.

Verdict

TRUE. $-|x| \leq x \sin \frac{1}{x} \leq |x|$ and $\pm|x| \to 0$, so by the Squeeze Theorem $\lim_{x \to 0} k(x) = 0 = k(0)$ (Question 5 of Lecture 2). Oscillation with a shrinking amplitude is continuous; oscillation with a fixed amplitude (Question 10) is not.

Being precise: continuity in the language of ε and δ

Limit at a (Lecture 3)

$\lim_{x \to a} f(x) = L$ means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$0 < |x - a| < \delta \;\Rightarrow\; |f(x) - L| < \varepsilon.$

Continuity at a

f continuous at a means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$|x - a| < \delta \;\Rightarrow\; |f(x) - f(a)| < \varepsilon.$

Being precise: continuity in the language of ε and δ

Limit at a (Lecture 3)

$\lim_{x \to a} f(x) = L$ means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$0 < |x - a| < \delta \;\Rightarrow\; |f(x) - L| < \varepsilon.$

Continuity at a

f continuous at a means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$|x - a| < \delta \;\Rightarrow\; |f(x) - f(a)| < \varepsilon.$

What this definition of continuity means

A small change in x makes only a small change in $f(x)$ — and the order is fixed: you say how close to $f(a)$ you want to be; I find how close to a you must stand.

Being precise: continuity in the language of ε and δ

Limit at a (Lecture 3)

$\lim_{x \to a} f(x) = L$ means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$0 < |x - a| < \delta \;\Rightarrow\; |f(x) - L| < \varepsilon.$

Continuity at a

f continuous at a means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$|x - a| < \delta \;\Rightarrow\; |f(x) - f(a)| < \varepsilon.$

What this definition of continuity means

A small change in x makes only a small change in $f(x)$ — and the order is fixed: you say how close to $f(a)$ you want to be; I find how close to a you must stand.

Two changes, and that is all:

Being precise: continuity in the language of ε and δ

Limit at a (Lecture 3)

$\lim_{x \to a} f(x) = L$ means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$0 < |x - a| < \delta \;\Rightarrow\; |f(x) - L| < \varepsilon.$

Continuity at a

f continuous at a means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$|x - a| < \delta \;\Rightarrow\; |f(x) - f(a)| < \varepsilon.$

What this definition of continuity means

A small change in x makes only a small change in $f(x)$ — and the order is fixed: you say how close to $f(a)$ you want to be; I find how close to a you must stand.

Two changes, and that is all:

1. The value $f(a)$ must exist.

Being precise: continuity in the language of ε and δ

Limit at a (Lecture 3)

$\lim_{x \to a} f(x) = L$ means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$0 < |x - a| < \delta \;\Rightarrow\; |f(x) - L| < \varepsilon.$

Continuity at a

f continuous at a means:

for every $\varepsilon > 0$ there is $\delta > 0$ with

$|x - a| < \delta \;\Rightarrow\; |f(x) - f(a)| < \varepsilon.$

What this definition of continuity means

A small change in x makes only a small change in $f(x)$ — and the order is fixed: you say how close to $f(a)$ you want to be; I find how close to a you must stand.

Two changes, and that is all:

1. The value $f(a)$ must exist.
2. The puncture $0 < |x - a|$ disappears — $x = a$ is let in.

A true story about your height

STAND UP

"At some moment of your life you were *exactly* 1 metre 50 tall."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

"At some moment of your life you were *exactly* 1 metre 50 tall."

Stand up if you think this is TRUE.

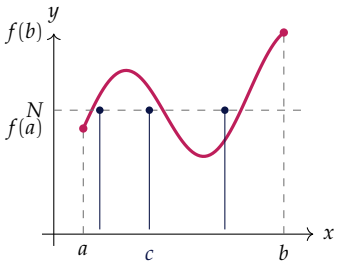
Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

TRUE (for everyone taller than 1.50 m today). You were about 50 cm long at birth, you are taller than 1.50 m now, and height does not jump — so at some instant it was exactly 1.50 m. That sentence is a theorem — the Intermediate Value Theorem — and it needs exactly one hypothesis: continuity. Here it is.

Intermediate Value Theorem

Suppose f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, $f(a) \neq f(b)$. Then there is a number c in (a, b) with $f(c) = N$.

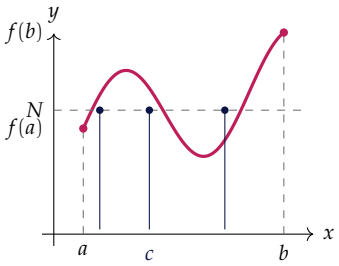


The Intermediate Value Theorem

Intermediate Value Theorem

Suppose f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, $f(a) \neq f(b)$. Then there is a number c in (a, b) with $f(c) = N$.

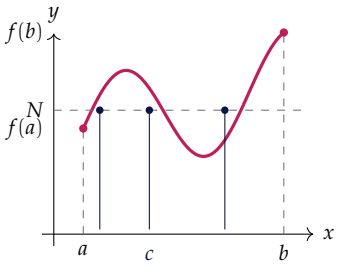
In words: a continuous function takes every value between two of its values — it cannot skip a level.



The Intermediate Value Theorem

Intermediate Value Theorem

Suppose f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, $f(a) \neq f(b)$. Then there is a number c in (a, b) with $f(c) = N$.



In words: a continuous function takes every value between two of its values — it cannot skip a level.

"There is a c" — at least one; the picture has three. The theorem does not say where.

The Intermediate Value Theorem

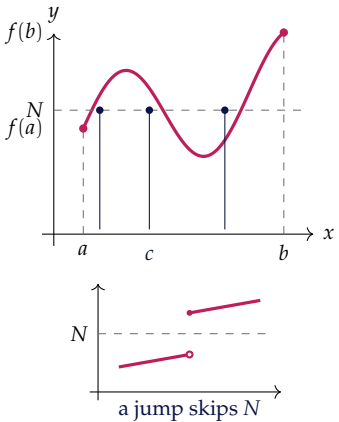
Intermediate Value Theorem

Suppose f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, $f(a) \neq f(b)$. Then there is a number c in (a, b) with $f(c) = N$.

In words: a continuous function takes every value between two of its values — it cannot skip a level.

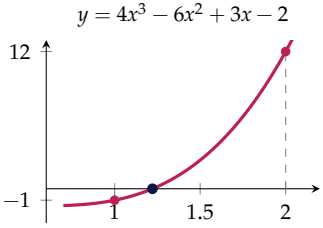
"There is a c" — at least one; the picture has three. The theorem does not say where.

The hypothesis is not decoration: a function with a jump can skip N (small picture). Without continuity the theorem is false.



Using the theorem: a root we cannot find

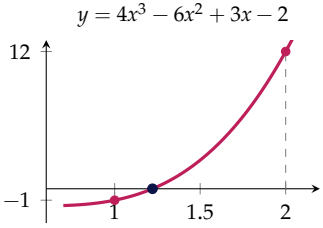
Claim. $4x^3 - 6x^2 + 3x - 2 = 0$ has a root between 1 and 2.



Using the theorem: a root we cannot find

Claim. $4x^3 - 6x^2 + 3x - 2 = 0$ has a root between 1 and 2.

Let $f(x) = 4x^3 - 6x^2 + 3x - 2$; we want c with $f(c) = 0$: take $N = 0$.

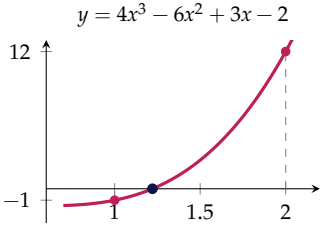


Using the theorem: a root we cannot find

Claim. $4x^3 - 6x^2 + 3x - 2 = 0$ has a root between 1 and 2.

Let $f(x) = 4x^3 - 6x^2 + 3x - 2$; we want c with $f(c) = 0$: take $N = 0$.

$f(1) = 4 - 6 + 3 - 2 = -1 < 0$ and
$f(2) = 32 - 24 + 6 - 2 = 12 > 0$.



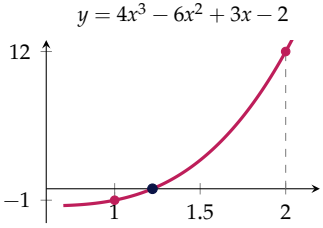
Using the theorem: a root we cannot find

Claim. $4x^3 - 6x^2 + 3x - 2 = 0$ has a root between 1 and 2.

Let $f(x) = 4x^3 - 6x^2 + 3x - 2$; we want c with $f(c) = 0$: take $N = 0$.

$f(1) = 4 - 6 + 3 - 2 = -1 < 0$ and
$f(2) = 32 - 24 + 6 - 2 = 12 > 0$.

f is a polynomial, hence continuous on $[1, 2]$, and 0 lies between $f(1)$ and $f(2)$.



Using the theorem: a root we cannot find

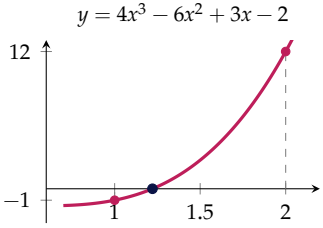
Claim. $4x^3 - 6x^2 + 3x - 2 = 0$ has a root between 1 and 2.

Let $f(x) = 4x^3 - 6x^2 + 3x - 2$; we want c with $f(c) = 0$: take $N = 0$.

$f(1) = 4 - 6 + 3 - 2 = -1 < 0$ and
$f(2) = 32 - 24 + 6 - 2 = 12 > 0$.

f is a polynomial, hence continuous on $[1, 2]$, and 0 lies between $f(1)$ and $f(2)$.

By the IVT there is $c \in (1, 2)$ with $f(c) = 0$: a root exists — we have not computed it.



Using the theorem: a root we cannot find

Claim. $4x^3 - 6x^2 + 3x - 2 = 0$ has a root between 1 and 2.

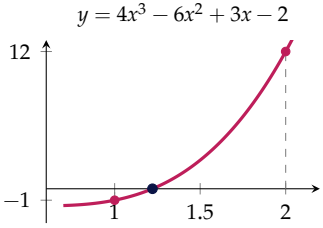
Let $f(x) = 4x^3 - 6x^2 + 3x - 2$; we want c with $f(c) = 0$: take $N = 0$.

$f(1) = 4 - 6 + 3 - 2 = -1 < 0$ and
$f(2) = 32 - 24 + 6 - 2 = 12 > 0$.

f is a polynomial, hence continuous on $[1, 2]$, and 0 lies between $f(1)$ and $f(2)$.

By the IVT there is $c \in (1, 2)$ with $f(c) = 0$: a root exists — we have not computed it.

Narrowing down: $f(1.2) = -0.128 < 0$, $f(1.3) = 0.548 > 0 \Rightarrow$ a root in $(1.2, 1.3)$. Repeat: this is how a calculator finds roots.



A little history · 1817

Everyone had always used the Intermediate Value Theorem — it looks obvious from the picture. Bernard Bolzano, a priest and mathematician in Prague, was the first to insist that "obvious" and "proved" are different things, and in 1817 he published a *purely analytic proof* of it, built on a definition of continuity very close to ours. Two years later he was dismissed from his chair for his sermons on peace and social justice, and did mathematics in isolation for the rest of his life; Cauchy proved the theorem independently in 1821. The lesson survived them both: a theorem is what you can *prove*, not what you can see.

Question 11

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.

B. The temperature was rising the whole time.

C. At 9 a.m. the temperature was 22 °C.

D. None of these must be true.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.

B. The temperature was rising the whole time.

C. At 9 a.m. the temperature was 22 °C.

D. None of these must be true.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

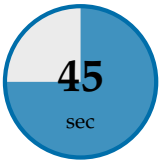
At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.

B. The temperature was rising the whole time.

C. At 9 a.m. the temperature was 22 °C.

D. None of these must be true.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

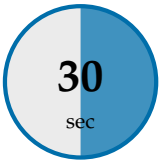
At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.

B. The temperature was rising the whole time.

C. At 9 a.m. the temperature was 22 °C.

D. None of these must be true.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

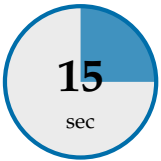
At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.

B. The temperature was rising the whole time.

C. At 9 a.m. the temperature was 22 °C.

D. None of these must be true.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.

B. The temperature was rising the whole time.

C. At 9 a.m. the temperature was 22 °C.

D. None of these must be true.

0
sec

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.
B. The temperature was rising the whole time.
C. At 9 a.m. the temperature was 22 °C.
D. None of these must be true.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.
B. The temperature was rising the whole time.
C. At 9 a.m. the temperature was 22 °C.
D. None of these must be true.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.
B. The temperature was rising the whole time.
C. At 9 a.m. the temperature was 22 °C.
D. None of these must be true.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.
B. The temperature was rising the whole time.
C. At 9 a.m. the temperature was 22 °C.
D. None of these must be true.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.
B. The temperature was rising the whole time.
C. At 9 a.m. the temperature was 22 °C.
D. None of these must be true.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.
B. The temperature was rising the whole time.
C. At 9 a.m. the temperature was 22 °C.
D. None of these must be true.

Hands down. Thank you.

A B C D

Question 11

EXPLAIN

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.

B. The temperature was rising the whole time.

C. At 9 a.m. the temperature was 22 °C.

D. None of these must be true.

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Question 11

EXPLAIN

At 6 a.m. the temperature in Tashkent was 14 °C; at noon it was 30 °C. Which statement must be true?

A. At some moment between 6 and 12 the temperature was exactly 20 °C.

B. The temperature was rising the whole time.

C. At 9 a.m. the temperature was 22 °C.

D. None of these must be true.

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

A. Temperature is a continuous function of time, and 20 lies between 14 and 30: the Intermediate Value Theorem with $N = 20$. B confuses "went up overall" with "went up all the time"; C is the "everything is linear" reflex — nothing says the rise was steady. D is the honest sceptic's answer, and the theorem is the reply to it.

STAND UP

"$f(2) = 5$ and $f(4) = -3$, so f has a zero somewhere between 2 and 4."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

"$f(2) = 5$ and $f(4) = -3$, so f has a zero somewhere between 2 and 4."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

FALSE — unless f is continuous. Take $f(x) = 5$ for $x \leq 3$ and $f(x) = -3$ for $x > 3$: it jumps over 0 and has no zero at all. Theorems have hypotheses for a reason. On the exam, write the sentence: "f is continuous on $[2, 4]$ (a polynomial), so by the IVT ..."

A surprising consequence

Claim. Right now there is at least one pair of diametrically opposite points on the equator with exactly the same temperature.

A surprising consequence

Claim. Right now there is at least one pair of diametrically opposite points on the equator with exactly the same temperature.

Let $T(\theta)$ be the temperature at longitude θ on the equator — a continuous function of θ. Set $g(\theta) = T(\theta) - T(\theta + 180^\circ)$.

A surprising consequence

Claim. Right now there is at least one pair of diametrically opposite points on the equator with exactly the same temperature.

Let $T(\theta)$ be the temperature at longitude θ on the equator — a continuous function of θ. Set $g(\theta) = T(\theta) - T(\theta + 180^\circ)$.

Then $g(180^\circ) = T(180^\circ) - T(360^\circ) = T(180^\circ) - T(0^\circ) = -g(0^\circ)$: $g(0^\circ)$ and $g(180^\circ)$ have opposite signs (or both are 0).

A surprising consequence

Claim. Right now there is at least one pair of diametrically opposite points on the equator with exactly the same temperature.

Let $T(\theta)$ be the temperature at longitude θ on the equator — a continuous function of θ. Set $g(\theta) = T(\theta) - T(\theta + 180^\circ)$.

Then $g(180^\circ) = T(180^\circ) - T(360^\circ) = T(180^\circ) - T(0^\circ) = -g(0^\circ)$: $g(0^\circ)$ and $g(180^\circ)$ have opposite signs (or both are 0).

g is continuous, so by the IVT there is a c between 0° and 180° with $g(c) = 0$, i.e. $T(c) = T(c + 180^\circ)$. $\square$

A surprising consequence

Claim. Right now there is at least one pair of diametrically opposite points on the equator with exactly the same temperature.

Let $T(\theta)$ be the temperature at longitude θ on the equator — a continuous function of θ. Set $g(\theta) = T(\theta) - T(\theta + 180^\circ)$.

Then $g(180^\circ) = T(180^\circ) - T(360^\circ) = T(180^\circ) - T(0^\circ) = -g(0^\circ)$: $g(0^\circ)$ and $g(180^\circ)$ have opposite signs (or both are 0).

g is continuous, so by the IVT there is a c between 0° and 180° with $g(c) = 0$, i.e. $T(c) = T(c + 180^\circ)$. $\square$

We know such a pair exists; we have no idea where it is. That is what an existence theorem does — and it is enough to stop a thermometer salesman who claims otherwise.

Today in three lines

1. Sums, differences, products, quotients (denominator $\neq 0$) and compositions of continuous functions are continuous — so anything built from polynomials, roots and trigonometric functions by these operations is continuous on its domain.

Today in three lines

1. Sums, differences, products, quotients (denominator $\neq 0$) and compositions of continuous functions are continuous — so anything built from polynomials, roots and trigonometric functions by these operations is continuous on its domain.
2. A limit passes inside a continuous function: $\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right)$.

Today in three lines

1. Sums, differences, products, quotients (denominator $\neq 0$) and compositions of continuous functions are continuous — so anything built from polynomials, roots and trigonometric functions by these operations is continuous on its domain.
2. A limit passes inside a continuous function: $\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right)$.
3. Intermediate Value Theorem: continuous on $[a, b] \Rightarrow$ every value between $f(a)$ and $f(b)$ is taken. The tool for "show that a root exists".

Today in three lines

1. Sums, differences, products, quotients (denominator $\neq 0$) and compositions of continuous functions are continuous — so anything built from polynomials, roots and trigonometric functions by these operations is continuous on its domain.
2. A limit passes inside a continuous function: $\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right)$.
3. Intermediate Value Theorem: continuous on $[a, b] \Rightarrow$ every value between $f(a)$ and $f(b)$ is taken. The tool for "show that a root exists".

This week's quiz question comes from Questions 8–13

Homework: Stewart §1.8, read and work through examples.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"The Intermediate Value Theorem needs continuity because …"*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.